Analysis of MEG data using deep learning techniques

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***Abstract*—This report investigates the application of deep learning techniques, specifically employing a Convolutional Neural Network (CNN), for the analysis of Magnetoencephalography (MEG) data. The focus lies in decoding patterns within MEG recordings to gain insights into the spatial and temporal dynamics of neural activity. This study aims to contribute valuable insights to the broader field of neuroscientific research and set the groundwork for future investigations into the application of machine learning procedures on brain activity data.**

List of Abbreviations

CNN; Convolutional Neural Network, RNN; Recurrent Neural Network, SVM; Support Vector Machine, KNN; K Nearest Neighbours, LSTM; Long Short-Term Memory, MEG; Magnetoencephalography, fMRI; functional magnetic resonance imaging, EEG; Electroencephalography, ReLU; Rectified linear unit

I. Introduction

In recent years, advancements in neuroscience and technology have enabled deeper exploration into the complexities of the human brain. One such advancement is Magnetoencephalography (MEG), a neuroimaging technique that measures the magnetic fields produced by neuronal activity. Utilizing multiple sensors placed on the scalp, MEG provides valuable insights, particularly in understanding mental disorders [10]. However, analyzing MEG data presents significant challenges due to its high dimensionality and the intricate patterns it contains.

This project leverages the power of deep learning, specifically a multi-layer CNN, to address these challenges. CNNs have shown remarkable success in processing and interpreting complex data structures, making them an ideal choice for this task. Our approach involves training the CNN to classify brain states from MEG data that correspond to four different tasks performed by the experiment subjects, an endeavor that could have profound implications in neuroscience research.

In this paper, we delve into the intricacies of MEG data, the methodology behind our deep learning model, and the technical choices we made. We aim to demonstrate the effectiveness of CNNs in classifying brain states from MEG data, thereby contributing to the broader field of neuroscience and computational brain research.

II. Related Work

In this section, we review relevant studies that inspired our work and have employed machine learning models (trained on MEG data) for diagnostic purposes.

There are various methods that have been developed and introduced to record and interpret the electrical activity of the brain, e.g., magnetoencephalography (MEG), functional magnetic resonance imaging (fMRI) and cortical evoked potentials monitored with electroencephalography (EEG). In our research, we only make use of MEG data to train our models as MEG is an imaging technique that is less prone to signal distortion created by brain tissues [3]. In recent years, the application of machine learning in neuroscientific research has gained momentum, with studies exploring its diagnostic potential in various neurological conditions. In [4], the researchers proved that a trained a Random Forest classifier on MEG data is able to discriminate different types of dementia e.g. Alzheimer's disease. In [5], the authors provide an extensive review of machine learning approaches, such as Support Vector Machines (SVM), K-Nearest Neighbors (KNN) and Convolutional neural networks (CNN) rained on MEG & EEG data for autism spectrum disorder classification. Finally, in [6], the reasearchers proposed a 3-layer CNN for motor task classification with MEG data from the BCI IV 3 dataset, that outperformed previous reasearcher's models (SVM, LDA, Linear SVM) significantly on the same dataset in both within-subject and cross-subject experiments. Although we are using a different dataset, and our models are used for broader classification tasks, this paper serves as an important reference in the broader context of our research, as it emphasized on the potential of CNNs for extracting meaningful spatial and temporal features from MEG data for efficient brain state classification. We hope that our work contributes to this

growing body of literature, particularly in the context of state classification on MEG data.

III. Approach

A. Dataset exploration

Generating a MEG dataset is a relatively expensive process due to the sophisticated scientific equipment used and the necessity to recruit participants. In order to overcome this issue, multiple researches use open-source datasets. Some of the most important public MEG datasets include ConnectomeDB based on the human connectome project (HCP) [7], the open MEG archive (OMEGA) from the McGill University, Cam-CAN dataset from The Cambridge Center for Aging and Neuroscience, A 204-subject multimodal neuroimaging dataset from the Radboud University [8], and many more. In this research endeavour we used a dataset provided by Dr. S. Mehrkanoon in the context of the Pattern Recognition and Deep Learning course of the University of Utrecht.

The dataset used in this research consists of MEG data from multiple people, performing four distinct tasks. The tasks performed were Resting tasks, Math & Story tasks, Working tasks and Motor tasks and are equally presented in our dataset. The sensors were able to record samples with a rate of 2034 Hz, meaning that during one second 2034 samples were recorded. The values recorded were measured in Tesla and the order of magnitude was in femtoTesla (10e-15 Tesla). In Figure 4 we can see a small subset of a MEG recording where the subject was performing resting tasks.

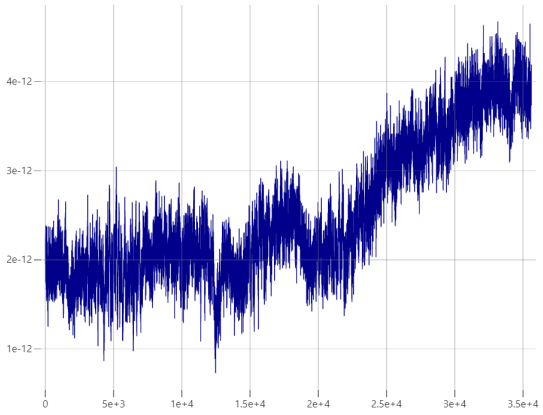


Fig. 1. Sample of Resting tasks MEG scan.

Due to the fact that our dataset included data from multiple subjects, we were able to perform two types of classification, intra and cross. Meaning that we were able to classify a MEG recording of a subject based on their previous recordings (intra) and based on recordings from different subjects (cross).

B. Data Preprocessing

1) Z-Score Normalization: The preprocessing of the Magnetoencephalography (MEG) dataset began with Z-score normalization, crucial for neural networks. This method was applied across all data files for each of the 248 features. By adjusting the values to a mean of zero and a standard deviation of one for each feature, we ensure a consistent scale across the dataset.

This feature-wise normalization is essential in MEG data processing, as it aligns different features on a common scale, enhancing the model's ability to learn from these features effectively. Applying normalization to each of the 248 features across all files helps in reducing biases caused by varied scales and facilitates faster convergence during the training of deep learning models.

2) Label Extraction: An essential step was the extraction and numerical encoding of labels representing different brain states. Each data segment was annotated with labels like rest, math, memory, or motor tasks, then converted into a numerical format following one-hot encoding. This conversion enabled the supervised learning process by providing a machine-readable format for categorical labels.

3) Downsampling: To address the high volume and complexity of the MEG data, downsampling was employed. This technique involved reducing the temporal resolution of the data by systematically selecting specific intervals. The purpose of downsampling in this context was to make the dataset more manageable for processing and analysis, ensuring computational efficiency.

C. Choice of model

We selected a multi-layer Convolutional Neural Network (CNN) to analyze the Magnetoencephalography (MEG) data. This decision is anchored in the proven efficacy of CNNs in processing high-dimensional data and their capacity to extract complex patterns, particularly relevant in datasets featuring spatial and temporal dimensions, like MEG recordings.

CNNs are known for their ability to learn hierarchical feature representations. This attribute is particularly valuable for deciphering patterns inherent in brain activity, as recorded by MEG. Such capabilities of CNNs have been demonstrated in a variety of applications, particularly in fields involving image and signal processing, which closely parallels the nature of MEG data analysis [1].

The layered architecture of CNNs, which typically includes convolutional and pooling layers, aligns well with the requirements for processing the spatial and temporal dynamics present in MEG data. By employing this model, we aim to leverage its strengths in feature extraction and pattern recognition to accurately classify different brain states based on MEG signals.

The choice of a CNN for this research is rooted in its compatibility with the nature of MEG data, its demonstrated effectiveness in similar tasks, and its capability to generalize from complex, high-dimensional datasets. Studies have shown CNNs to be highly effective in classifying and analyzing MEG data, further validating our choice of model for this study [1] [2].

D. Model Architecture

The model architecture employed in this research is a Sequential model utilizing Convolutional Neural Networks,

specifically tailored for processing our dataset. This architecture effectively captures the spatial hierarchies in the data, making it particularly suitable for our analysis.

1) Convolutional Layers: The architecture begins with a Conv1D layer comprising 32 filters and a kernel size of 3, using the ReLU activation function to extract initial features from the input data, which is configured to match our custom data dimensions. This is immediately followed by a MaxPooling1D layer with a pool size of 2, aiming to reduce the dimensionality and enhance the robustness of the features extracted. The model then repeats this configuration, adding another Conv1D layer with 64 filters to learn more complex features, and another MaxPooling1D layer for further dimensionality reduction and feature extraction.

2) Flattening and Dense Layers: Post the convolutional and pooling layers, the model employs a Flatten operation, transforming the data into a single vector form, making it suitable for processing via dense layers. This is followed by a dense layer with 128 units, using the ReLU activation function, which interprets the features extracted by the convolutional layers. The architecture culminates with a final dense layer comprising 4 units and utilizing the softmax activation function, designed to output a probability distribution over the four classes in our classification task.

3) Compilation: The model is compiled using the Adam optimizer, selected for its effectiveness in managing sparse gradients and its adaptive learning rate capabilities, coupled with the categorical crossentropy loss function, a standard choice for multi-class classification tasks. The learning rate is set to 0.001. To monitor and evaluate the model's performance during training, the 'accuracy' metric is tracked.

This architecture strikes a balance between computational efficiency and the capacity to capture complex patterns within the dataset. It is finely tuned to meet the specific demands of our dataset and the nature of the classification task, ensuring an optimal blend of performance and practicality within the ambit of our research.

IV. Results

A. Intra-Dataset Classification

For the intra-dataset classification, where the model is trained and tested on data from the same subjects, both RNN and CNN architectures achieved accuracy of 100%. This suggests that both RNN and CNN are highly effective in capturing the temporal patterns in the MEG data corresponding to different brain states.

The LSTM architecture, didn't achieve as high results but exhibited a slightly lower accuracy of 66% in this scenario. This may be attributed to the specific characteristics of the MEG data, where the temporal dependencies captured by LSTM might not be as crucial as the spatial features emphasized by the other architectures.

B. Cross-Dataset Classification

In the cross-dataset classification, where the model is trained on data from one set of subjects and tested on data from a different set, the results demonstrate the strength of the CNN architecture. The CNN achieved an accuracy of 87.50% over the course of 10 epochs of training, showcasing its ability to generalize well to unseen subjects. On the other hand, both the RNN and LSTM architectures exhibited varying accuracies in the cross-dataset scenario. The RNN achieved an accuracy of 75%, while the LSTM, with an accuracy of 57%. This suggests that the temporal dependencies captured by RNN and LSTM architectures may not generalize as effectively across different subjects compared to the features emphasized by the CNN.

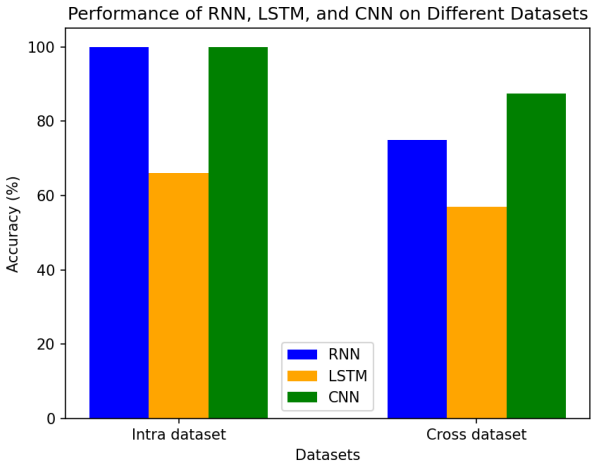


Fig. 2. Performance of RNN, LSTM & CNN over Intra and Cross datasets

C. CNN Hyperparameters Tuning

The architecture's initial parameters were systematically determined through a grid search process. As shown in figure 3, the Conv1D layer was configured with 32 filters and a kernel size of 3, employing the ReLU activation function to extract essential features from the input data. Subsequently, a MaxPooling1D layer with a pool size of 2 was applied to reduce dimensionality and enhance the robustness of the extracted features. This configuration was iteratively replicated with an additional Conv1D layer featuring 64 filters, allowing the model to learn more complex features. The subsequent MaxPooling1D layer further contributed to dimensionality reduction and feature extraction. The grid search methodology facilitated the exploration of the model's parameter combinations to identify the optimal configuration for the given MEG dataset.

The model is compiled using the Adam optimizer with a learning rate of 0.001, and categorical cross-entropy is employed as the loss function. The Adam optimizer's [9] ability to navigate diverse landscapes of loss functions, in addition to its computational efficiency, makes it a well-established and very often preferred choice for optimizing CNNs, contributing significantly to the success of deep learning models different processing tasks.

The down-sampling factor of 2 played a crucial role in enhancing the model's efficiency by reducing the temporal

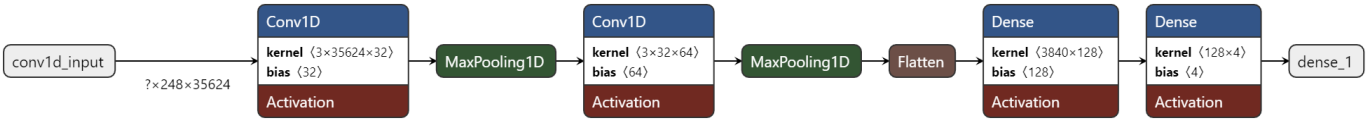


Fig. 3. CNN model architecture

resolution of the data. This not only accelerated the training process but also managed to maintain essential information for accurate classification, as evidenced by the results.

Experimentation revealed that the best setup for the MEG dataset was a batch size of 1. In scenarios where the number of samples used in each iteration to update the model's weights is set to 1, the model demonstrated better accuracy. This finding implies that smaller batch sizes are more beneficial in this context, potentially due to the limited dataset. When using larger batch sizes, the model's accuracy dropped below the 65%.

1) Disparity between accuracies: During the initial training phase of our CNN model, a significant discrepancy was observed between the training and testing accuracies. The model achieved a perfect training accuracy of 100%, while the testing accuracy lagged at 75%. This substantial gap suggests a case of overfitting, where the model memorizes training data patterns but fails to generalize these patterns to unseen data. Several factors can contribute to this phenomenon, including model complexity, inadequate training data, and lack of regularization.

2) Addressing the disparity: To mitigate overfitting and improve the model's generalization capability, we incorporated regularization techniques into our architecture. Specifically, we introduced a dropout layer. Dropout is a widely used regularization technique that randomly deactivates a fraction of neurons during the training process, preventing the model from becoming overly reliant on specific weights and thus promoting a more distributed and robust feature learning. After applying dropout and experimenting with different values, we found out that a dropout rate of 0.25 gave the best results. The model's testing accuracy improved to 87.5%, significantly narrowing the gap between training and testing accuracies and indicating enhanced generalization.

The hyperparameter choices were informed by a systematic exploration of different configurations. Through thorough testing and experimentation, we were able to identify the hyperparameters that performed the best in both Intra and Cross dataset classifications.

D. Cross CNN Confusion Matrix

The confusion matrix for the classification of brain states using our CNN demonstrates a high level of model accuracy for 'Resting', 'Math & Story', and 'Working Memory' states, with each of these categories achieving 100% accuracy in classification. This indicates that the CNN has effectively learned the distinct features of these brain states, leading to a precise identification without any mislabeling.

On the other hand, the classification for the 'Motor' state shows some misclassification, with a 75% accuracy rate and the remaining 25% being wrongly classified as 'Working Memory'. The result suggests that while the model is able to recognize 'Motor' state features with a high degree of accuracy, there is an area of confusion between 'Motor' and 'Working Memory' states that is caused by the similarity in their neural signatures as captured by the MEG data.

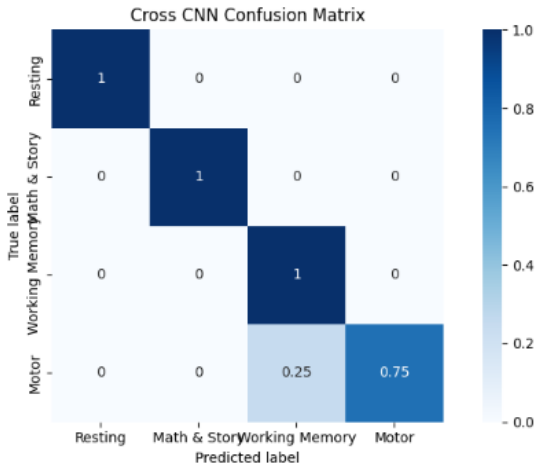


Fig. 4. Confusion matrix of the four classes in cross evaluation

V. Conclusion

Our research focused on identifying the most effective deep learning technique for understanding MEG data. With 100% accuracy in intra-dataset classification and 87.50% in cross-dataset classification, the CNN emerged as a robust solution for finding patterns in both spatial and temporal dynamics. The adopted architecture, complemented by the preprocessing techniques of Z-Score Normalization and downsampling, achieved a harmonious balance between computational efficiency and pattern extraction. The study not only showcased high classification accuracy but also provided insights into the effectiveness of CNNs for MEG data analysis. The careful consideration of hyperparameters contributed to the model's ability to process and classify MEG data. Our research sets a foundation for future investigations into refining model architectures, exploring larger datasets, and possibly extending applications beyond classification tasks. Moving forward, our focus should shift towards refining model architectures and exploring larger datasets to deepen insights and propel the application of MEG and other types of brain activity data in neuroscientific research.

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